norm : I.Tm A → Nf A

⌜_⌝

Nf Bool = 𝟚
Nf Nat = ℕ

teljesség : ⌜norm t⌝ = t
stab : norm ⌜n⌝ = n

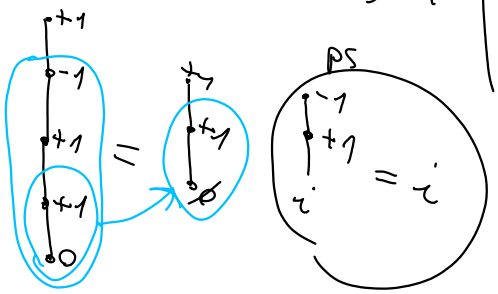
A szabad monoid ⇄ List A

Tm : Set
ι : A → Tm
_+_ : Tm → Tm → Tm
∅ : Tm
az
idl : ∅ + t = t
idr

egyéb számok

Tm : Set
zero : Tm
suc : Tm → Tm
pred : Tm → Tm
ps : pred (suc i) = i
sp : suc (pred i) = i

Nf
(ℕ ⊎ T) ⊎ ℕ



norm (suc (pred (suc (suc zero))) ...) = ι_2 1

ι_1 : A → A ⊎ B
ι_2 : B → A ⊎ B

norm (suc (pred zero)) = ι_1 (ι_2 triv)

ite true u v = u
ite false u v = v

ite t u u = u

pl.

t := nothing
u, v := just tt

Ty = Set
Tm A = Maybe A
Bool = 𝟚
Nat = ℕ
true = just tt
false = just ff
ite nothing u v := nothing
ite (just tt) u v := u
ff := v

Maybe A ≅ A ⊎ T
nothing : Maybe A
just : A → Maybe A

M

M.⟦_⟧ : I.Tm Bool → M.Tm (𝟚) = Maybe 𝟚

u = ite true u v = ite false u v = v
$ite\beta_1$ $ite\beta_2$

true = false

zero n = zero m

$isZero\beta_1$ $isZero\beta_1$
true = isZero (zero 0) = isZero (zero 1) = false

Def | lokális definíció:

(def $\otimes := \underbrace{m1 \pm m2}_{:\mathrm{Tm}\,A}$ in $(\otimes + \otimes) = (m1 + m2) + (m1 + m2)$

let $\qquad\qquad\qquad

elsőrendű pred.

$$\mathrm{def} : \mathrm{Tm}\ A \to (\underline{\mathrm{Tm}\ A} \to \mathrm{Tm}\ B) \to \mathrm{Tm}\ B$$

változók $\to \Gamma$ környezete

$$\frac{t : \mathrm{Tm}\ \Gamma\ A \qquad u : \mathrm{Tm}\ (\Gamma, x{:}A)\ B}{\mathrm{def}\ t\ u : \mathrm{Tm}\ \Gamma\ B}$$

$$\mathrm{def} : \mathrm{Tm}\ \Gamma\ A \to \mathrm{Tm}\ (\Gamma, x{:}A)\ B \to \mathrm{Tm}\ \Gamma\ B$$

$$\mathrm{def}\ x := 2\ \mathrm{in}\ \mathrm{def}\ y := x+3\ \mathrm{in}\ x+y = \mathrm{def}\ y := 2+3\ \mathrm{in}\ 2+y$$

$$\| $$

$$2 + (2+3) = 7$$

$$\frac{\dfrac{x : \mathrm{Tm}\ (x{:}\mathrm{Nat})\ \mathrm{Nat} \qquad 3 : \mathrm{Tm}\ (x{:}\mathrm{Nat})\ \mathrm{Nat}}{x+3 : \mathrm{Tm}\ (x{:}\mathrm{Nat})\ \mathrm{Nat}} \qquad \dfrac{x+y : \mathrm{Tm}\ (x{:}\mathrm{Nat}, y{:}\mathrm{Nat})\ \mathrm{Nat}}{\mathrm{def}\ y := x+3\ \mathrm{in}\ x+y : \mathrm{Tm}\ (\circ, x{:}\mathrm{Nat})\ \mathrm{Nat}}}{\mathrm{def}\ x := 2\ \mathrm{in}\ \mathrm{def}\ y := x+3\ \mathrm{in}\ x+y : \mathrm{Tm}\ \circ\ \mathrm{Nat}}\ \left(2 : \mathrm{Tm}\ \circ\ \mathrm{Nat}\right)$$

DefAST $\to$

DefWT

Def

$\mathrm{Tm} : \mathrm{Set}$

$_+_ : \mathrm{Tm} \to \mathrm{Tm} \to \mathrm{Tm}$

$\mathrm{true} : \mathrm{Tm}$

....

$\mathrm{def} : \mathrm{String} \to \mathrm{Tm} \to \mathrm{Tm} \to \mathrm{Tm}$

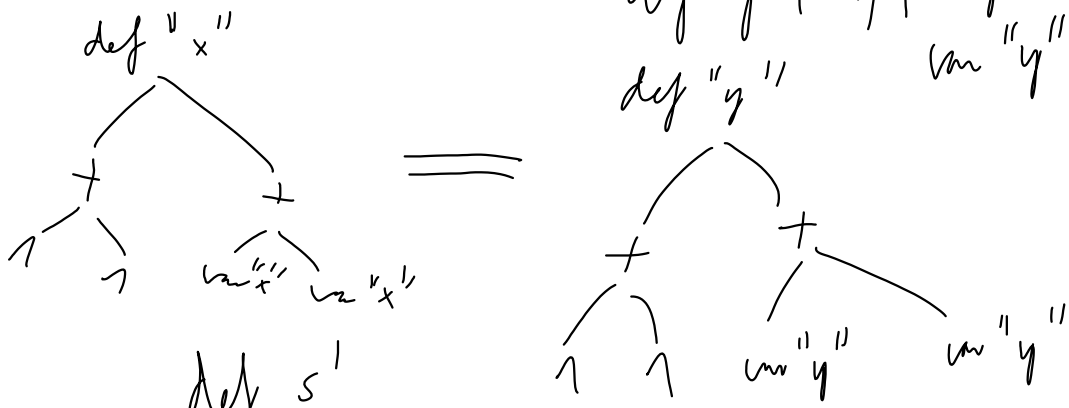
$\mathrm{var} : \mathrm{String} \to \mathrm{Tm}$

$\mathrm{def}\ x := 1+2\ \mathrm{in}\ x+x$

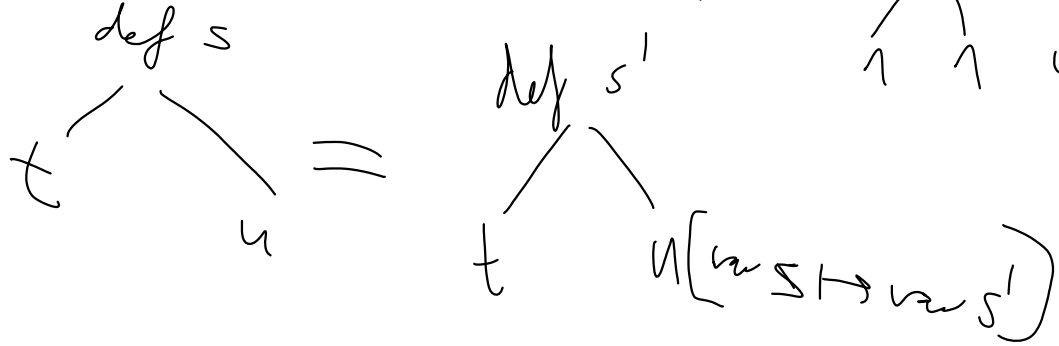
$\mathrm{def}\ \text{"x"}\ (1+1)\ (\mathrm{var}\ \text{"x"} + \mathrm{var}\ \text{"y"})$

$\mathrm{def}\ \text{"y"}\ (1+1)\ (\mathrm{var}\ \text{"y"} + \mathrm{var}\ \text{"y"})$

DefABT

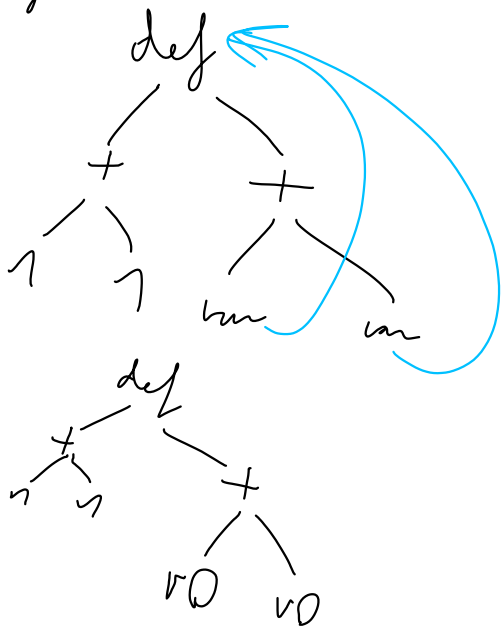


ált.:



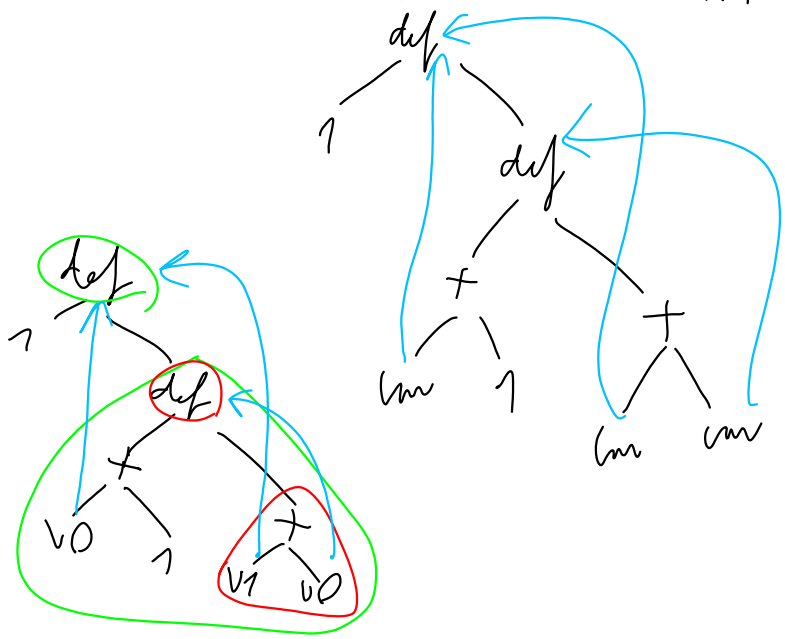
"kötött változó neve nem számít"

def x:= 1+1 in var "x" + var "x"



De Bruijn

def x:=1 in def y:=x+1 in x+y



v0, v1: De Bruijn-indexek

def x:=true in istero (def y:=m 2 in ite x (istero y) (m 3))

def true (istero (def (m 2) (ite v1 (istero v0) (m 3))))

DefABT

nincs név

Tm : ℕ → Set

Var : ℕ → Set

vz : Var (1+n)

vs : Var n → Var (1+n)

var : Var n → Tm n

true, false : Tm n

ite : Tm n → Tm n → Tm n → Tm n

m : ℕ → Tm n

_+_ : Tm n → Tm n → Tm n

istero : Tm n → Tm n

def : Tm n → Tm (1+n) → Tm n

a szabad változók maximális száma

Var n = {v0, v1, ..., v(n-1)}

v0 = vz

v1 = vs vz

v2 = vs (vs vz)

v3 = vs (vs (vs vz))

....

def x:=m 1 in x+x ← zárt

$$\dfrac{\dfrac{}{\mathrm{nu}\ 1 : \mathrm{Tm}\ 0} \qquad \dfrac{\dfrac{\dfrac{}{\mathrm{vz} : \mathrm{Var}\ 1}}{\mathrm{var}\ \mathrm{vz} : \mathrm{Tm}\ 1} \qquad \dfrac{\dfrac{}{\mathrm{vz} : \mathrm{Var}\ 1}}{\mathrm{var}\ \mathrm{vz} : \mathrm{Tm}\ 1}}{\mathrm{var}\ \mathrm{vz} + \mathrm{var}\ \mathrm{vz} : \mathrm{Tm}\ 1}}{\mathrm{def}\ (\mathrm{nu}\ 1)\ (\mathrm{var}\ \mathrm{vz} + \mathrm{var}\ \mathrm{vz}) : \mathrm{Tm}\ 0}$$

$$\mathrm{def}\ x := \mathrm{nu}\ 1 + y\ \mathrm{in}\ \mathrm{def}\ z := x + y\ \mathrm{in}\ x + (y + z)$$

$$\dfrac{\dfrac{\dfrac{\dfrac{}{\mathrm{vz} : \mathrm{Var}\ 2}}{\mathrm{v0} : \mathrm{Tm}\ 2} \qquad \dfrac{\dfrac{\dfrac{}{\mathrm{vz} : \mathrm{Var}\ 1}}{\mathrm{vs}\ \mathrm{vz} : \mathrm{Var}\ 2}}{\mathrm{v1} : \mathrm{Tm}\ 2}}{\mathrm{v0} + \mathrm{v1} : \mathrm{Tm}\ 2} \qquad \dfrac{\dfrac{\dfrac{\dfrac{}{\mathrm{vz} : \mathrm{Var}\ 1}}{\mathrm{vs}\ \mathrm{vz} : \mathrm{Var}\ 2}}{\mathrm{v1} : \mathrm{Tm}\ 2} \qquad \dfrac{\dfrac{\dfrac{\dfrac{}{\mathrm{vz} : \mathrm{Var}\ 1}}{\mathrm{vs}\ \mathrm{vz} : \mathrm{Var}\ 2}}{\mathrm{vs}\ (\mathrm{vs}\ \mathrm{vz}) : \mathrm{Var}\ 3} \qquad \vdots}{\mathrm{v2} : \mathrm{Tm}\ 3} \qquad \mathrm{v0} : \mathrm{Tm}\ 3}{\mathrm{v2} + \mathrm{v0} : \mathrm{Tm}\ 3}}{\mathrm{v1} + (\mathrm{v2} + \mathrm{v0}) : \mathrm{Tm}\ 3}}{\mathrm{def}\ (\mathrm{var}\ \mathrm{vz} + \mathrm{var}\ (\mathrm{vs}\ \mathrm{vz}))\ (\mathrm{v1} + (\mathrm{v2} + \mathrm{v0})) : \mathrm{Tm}\ 2}$$

$$\dfrac{\dfrac{}{\mathrm{nu}\ 1 + \mathrm{var}\ \mathrm{vz} : \mathrm{Tm}\ 1} \qquad \mathrm{def}\ (\mathrm{var}\ \mathrm{vz} + \mathrm{var}\ (\mathrm{vs}\ \mathrm{vz}))\ (\mathrm{v1} + (\mathrm{v2} + \mathrm{v0})) : \mathrm{Tm}\ 2}{\mathrm{def}\ (\mathrm{nu}\ 1 + \mathrm{var}\ \mathrm{vz})\ (\mathrm{def}\ (\mathrm{var}\ \mathrm{vz} + \mathrm{var}\ (\mathrm{vs}\ \mathrm{vz}))\ (\mathrm{var}\ (\mathrm{vs}\ \mathrm{vz}) + (\mathrm{var}\ (\mathrm{vs}\ (\mathrm{vs}\ \mathrm{vz})) + \mathrm{var}\ \mathrm{vz}))) : \mathrm{Tm}\ 1}$$

kov. óra 11 percel korábbi